6156 PROJECT PROPOSAL

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# Predicting Measles Outbreak Risk Across U.S. Counties

# Goal: Identify High-Risk Counties Before Outbreaks Occur

**Problem:** Public health reacts *after* outbreaks occur

**Impact:** Proactive targeting prevents outbreaks, saving ~$250k per outbreak

**Approach:** Supervised ML predicts high-risk counties

**End users:** Public health agencies via interactive dashboard

# Early Detection Reduces Millions in Public Health Costs & Protect Lives

Initial outbreak = **$250k+**

Additional cases = **$15k each**

Preventing just **3 outbreaks = ~$3 Million and Protects Lives**

Model enables **targeted intervention**

# Model Output: County-Level Risk Scores and Classifications

| Each county receives | Output delivered as | Used for |
| --- | --- | --- |
| • **Risk probability (0-1)**<br>• Binary classification**: High Risk (1) or Low Risk (0)** | • **Interactive U.S. map**<br>• Highlights high-risk counties | • Targeted MMR vaccination campaigns<br>• Resource allocation decisions |

# Model Predicts County-Level Measles Outbreak Risk

Supervised classification model

- **Target**:1 if county has >2 measles cases, 0 otherwise

Data sources:

- Measles cases (Johns Hopkins)
- Vaccine Hesitancy (CDC)
- Demographics (Census)

Challenge: **Rare events (class imbalance)**

- 3147 counties in US
- **~8% of counties have measles cases**

**Goal: maximize recall (missing an outbreak is costly)**

# Dashboard Enables Targeted Public Health Action

Map shows:

- Current outbreak counties
- **Predicted at-risk counties**

Click a county → see:

- **Why it's at risk (SHAP)**
- Key drivers (vaccination, age, etc.)

Runs **weekly (batch updates)**

**Used for proactive vaccination and resource allocation decisions**

# Project Roadmap: From Data to Deployment

| Milestone | Key Tasks | Due Date |
|---|---|---|
| M1: Project Proposal | Data merging & cleaning | Wednesday, March 18th |
| M2: Feedback & Data Preparation | Feature engineering & EDA | Wednesday, March 25th |
| M3: Model Dev | Model development (focus on recall) | Wednesday, April 1st |
| M4: Risk & MRM Documentation | Risk & MRM documentation | Wednesday, April 8th |
| M5: Finalize & Polish | Final deliverable + presentation slides (includes buffer time) | Wednesday, April 22nd |

# Sources

1. Willens, Victoria. October 28, 2025. “Estimating the Financial Costs of Measles Outbreaks.” *Johns Hopkins, Bloomberg School of Public Health.* Estimating the Financial Costs of Measles Outbreaks | International Vaccine Access Center.

# Data Sources

- CDC Vaccine Hesitancy (county-level): https://data.cdc.gov/Vaccinations/Vaccine-Hesitancy-for-COVID-19-County-and-local-es/q9mh-h2tw
- Johns Hopkins Measles Tracker: U.S. Measles Tracker | International Vaccine Access Center
- CDC MMR Vaccination CoverageVaccination Coverage and Exemptions among Kindergartners | Data | Centers for Disease Control and Prevention
- US Census Demographics: County Population by Characteristics: 2020-2024